2.2D Die last Integrated Substrate for High Performance Applications

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***Abstract*—In this work, a new integrated substrate structure "2.2D" is demonstrated. 2.2D substrate is a true die last solution. This solution used thin film RDL directly bonded to the substrate. This simplifies the 2.5D structure and further reduces the cost and improves the product cycle time. In this work, the 2.2D test vehicle is demonstrated with two metal layers of thin film that is bonded to a ceramic substrate. Serpentine lines run from thin film RDL to the substrate were evaluated. The process developed in this study demonstrates good registration between thin film RDL and substrate. The 2.2D TV demonstrates good warpage behavior up to the solder reflow temperature. The 2.2D structure shows good potential to be used for high performance computing substrate.**

Keywords*— *2.0D, 2.1D, 2.2D, 2.3D, 2.5D, Silicon interposer, eHDF, CoWoS, RDL interposer, TSV-less, Solder-less, Integrated substrate, Heterogeneous Integration, Chiplet, High Performance computing.

I. Introduction

As society is embracing the age of AI, autonomous driving vehicles and smart-X (city, hospital, agriculture, etc.), the demand of high-performance computing power is ever increasing. Semiconductor is still following Moore's law by shrinking dimensions from 5nm, 3nm and 2nm and beyond. This drives up the need of interconnection density and of reducing the interconnection length between the connecting sites.

Traditional packaging cab be divided into different levels [1]. But due to the performance requirements, "level one" package (chip to substrate) is disappearing due to the benefit of reducing the connecting distance between chips by using bare dies. AMD's 2016 Fiji product uses bare dies for the interconnection between GPU and HBMs. This structure becomes the facto standard for the HPC packaging as shown in Fig. 1. In this case, 2.5D silicon interposers are used.

The term silicon interposer is interesting since silicon is only acting as the mechanical support of the fine line (down to submicron). Silicon manufacture has good infrastructure to meet the fine line requirements of the interposer. However, silicon is a semiconducting material that needs isolation of TSV (through silicon via). The TSV components can be modelled as resistance, capacitance and inductance [2]. Hence efforts have been made to cut the thickness of the silicon interposer or even to remove it completely. Various TSV-less solutions have been developed in the past. Examples are Amkor's SLIT [3], Samsung's RDL interposer [4] and TSMC's CoWoS-R [5].

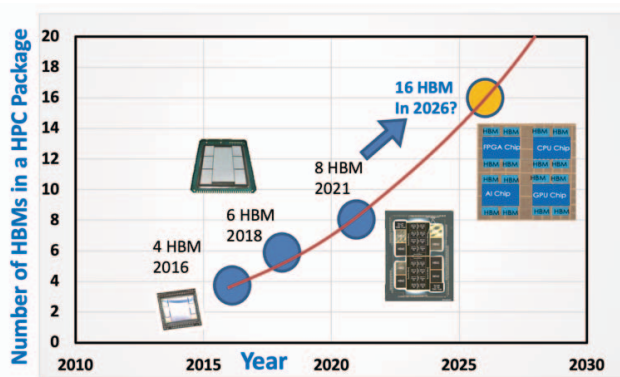

Fig. 1. The number of HBMs inside a substrate doubles every five years. Unofficial rule of HPC packaging.

Substrate with "coreless" structure has been in mass production over ten years due to the benefit of electrical performance of the system [6]. Solders are used for interconnection of silicon interposer and the underlying substrate. However, solder itself also can be modelled as resistance, capacitance and inductance [2]. Hence to remove the

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